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Identifying neuronal correlates of dying and resuscitation in a model of reversible brain anoxia

Short title: Neuronal activity from anoxia to resuscitation

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Highlights

- Anoxia-triggered silencing of ECoG is preceded by surges of fast then slow activities.
- Neocortical neurons at near-death undergo a massive membrane depolarizing shift.
- This neuronal anoxic depolarization is identifiable as a polyphasic wave in the ECoG.
- A timely re-oxygenation can rescue transmembrane potential and cellular excitability.
- A second ECoG wave marks the regaining of neuronal functions upon reanimation.

Abstract

We developed a new rodent model of reversible brain anoxia and performed continuous electrocorticographic (ECoG) and intracellular recordings of neocortical neurons to identify in real-time the cellular and network dynamics that successively emerge throughout the dying-to-recovery process. Along with a global decrease in ECoG amplitude, deprivation of oxygen supply resulted in an early

surge of beta-gamma activities, accompanied by rhythmic membrane depolarizations and regular firing in pyramidal neurons. ECoG and intracellular signals were then dominated by low-frequency activities which progressively declined towards isoelectric levels. Cortical neurons during the isoelectric state underwent a massive membrane potential depolarizing shift, captured in the ECoG as a large amplitude triphasic wave known as the “wave-of-death” (WoD). This neuronal anoxic depolarization, associated with a block of action potentials and a loss of cell integrative properties, could however be reversed if brain re-oxygenation was rapidly restored (within 2–3.5 min). The subsequent slow repolarization of neocortical neurons resulted in a second identifiable ECoG wave we termed “wave-of-resuscitation” since it inaugurated the progressive regaining of pre-anoxic synaptic and firing activities. These results demonstrate that the WoD is not a biomarker of an irremediable death and unveil the cellular correlates of a novel ECoG wave that may be predictive of a successful recovery. The identification of real-time biomarkers of onset and termination of cell anoxic insult could benefit research on interventional strategies to optimize resuscitation procedures.

Abbreviations: AP, action potential; CV, coefficient of variation; ECoG, electrocorticogram; EEG, electroencephalogram; EtCO₂, end-tidal carbon dioxide concentration; ISI, inter-spike interval; M1, primary motor cortex; S1, primary somatosensory cortex; SpO₂, arterial oxygen saturation; Vm = membrane potential; Rm, membrane resistance; τ_m , membrane time constant; WoD, wave-of-death; WoR, wave-of-resuscitation

Keywords: Brain anoxia; Dying; Resuscitation; Near-Death Experience; Neocortex; Neuronal Excitability

1. Introduction

Elucidating the neuronal events subtending the dying process during brain anoxia – the ultimate cause of death (Laureys, 2005) – and those occurring during the regaining of cerebral functions when a timely resuscitation procedure is initiated, is among the most challenging issues of clinical neurology and neurosciences.

Previous studies addressing this question in animal models of global brain anoxia have identified sequential changes in cortical electroencephalographic (EEG) patterns during the dying process. For example, inducing an abrupt cardiac arrest in rats by intra-cardiac injection of potassium chloride led to a fast decline in EEG amplitude associated, within the first 30 s preceding the isoelectric line, with a surge of widespread high-frequency gamma oscillations (Borjigin et al., 2013). In another cardiac arrest rodent model, wherein anoxia was induced by an interruption of the artificial ventilation, the attenuation of EEG activity towards the isoelectric state was accompanied by a decreased power in all frequency bands and an increased functional connectivity between bilateral frontal cortices (Lee et al., 2017). Interruption of cerebral blood flow by decapitation in rats also resulted in a rapid loss of global EEG power, occasionally associated with slow waves or low amplitude fast activity (Vanderwolf et al., 1988; van Rijn et al., 2011; Kongara et al., 2014). Around a minute after decapitation, during the electro-cerebral silence, a slow and large amplitude polyphasic wave was suddenly detected on the EEG. This wave was suggested to mark the synchronous and irreversible death of neurons and therefore named the “wave-of-death” (WoD) (van Rijn et al., 2011; see also Fig. 2 in Kaminogo et al., 1998). Based on a biophysical single-cell model, Zandt et al. (2011) proposed that the WoD would reflect the massive and simultaneous depolarization of a large number of neurons, caused by a cascade of oxygen deprivation-gated neuronal mechanisms initiated by the dysfunction of the ATP-dependent sodium-potassium pumps and the loss of neuronal transmembrane potential (Lipton, 1999; Shimoda and Polak, 2011).

Similar observations were made from critically-ill patients receiving palliative care. By means of EEG-derived multivariate indexes, a transient rise in activity, presumed to represent high-frequency brain signals, was identified within 1-2 min after cardiac arrest (Chawla et al., 2009, 2017; Auyong et al., 2010). This finding, suggesting the persistence in the dying brain of activity patterns generally associated with conscious waking state (Seth et al., 2005), led to the hypothesis that end-of-life surges of electrical activities could provide a neuronal basis to the near-death experiences (NDEs) (Chawla et al., 2009, 2017) – these vivid mental experiences reported by many cardiac arrest survivors (van Lommel et al., 2001). A propagating polyphasic EEG wave, considered as a terminal spreading depolarization analogous to the rodent WoD, was also recently detected during the dying process in

patients with devastating brain injury after withdrawal of life-sustaining treatments (Dreier et al., 2018).

The neuronal origins of these near-death brain electrical activities recorded *in vivo* remain unknown, since previous human and animal studies were limited to their observation in the EEG. In addition, the lethal protocols used in rodent models have precluded an examination of the neuronal recovery process during resuscitating procedures. Here, by combining simultaneous recordings of arterial blood gases, heart rate, electrocorticographic (ECoG) potentials, intracortical direct current (DC) shift and neocortical neurons intracellular activity in a rodent model of transient anoxia, we provide the first *in vivo* characterization of the cellular and synaptic changes subtending the ECoG dynamics that successively emerge throughout the anoxic period and subsequent resuscitation process. We also demonstrate that the anoxic depolarization of neocortical neurons causing the WoD can be reversed and identify a new relevant ECoG wave, the “wave-of-resuscitation” (WoR), which inaugurates the recovery of neuronal and network activities following a timely re-oxygenation of the brain.

2. Materials and methods

Experiments were carried out in accordance with the European Union guidelines (directive 2010/63/EU) and received approval (n° 2019051019017280) from the French Ministry for Research and the Charles Darwin Ethical Committee on Animal Experimentation (C2EA-05). Every precaution was taken to minimize stress, suffering and the number of animals used in each series of experiments.

2.1. Animals, surgery and assessment of physiological conditions

Adult male Sprague Dawley rats (n = 29; Charles River Laboratories), weighting 300–500 g, were initially anaesthetized by intraperitoneal injection of sodium pentobarbital (27 mg/kg; Centravet) and ketamine (25 mg/kg; Centravet) to perform the surgical procedures. The animals were then cannulated after incision of the trachea and placed in a stereotaxic frame. Incisions and pressure points were repeatedly infiltrated with lidocaine (2%; Centravet) and the core temperature was stabilized (36.5–37.5°C) throughout the experiment using a feedback-controlled heating blanket. For subsequent ECoG, extracellular and intracellular recordings, small craniotomies were made above left and right primary somatosensory (S1) cortices and left primary motor (M1) cortex. After surgery, animals were paralyzed by intramuscular injection of gallamine triethiodide (40 mg/2h; Sigma-Aldrich) and kept under artificial ventilation with room air at a rate of 85 cycles/min, a breath volume of 2.6 ml/cycle and an inspiratory pressure of 12 mbar. Anesthesia was then maintained during recordings by regular

injection of pentobarbital (11 mg/kg/h, ip). Pulse oximetry, capnography and cardiopulmonary parameters were continuously assessed with a veterinary vital signs monitor (LifeWindow unit, Digicare). During control periods, the arterial oxygen saturation (SpO₂) and end-tidal carbon dioxide concentration (EtCO₂) values ranged between 95–98% and 4–5.5 %, respectively, and the heart rate was of 350–450 beats/min (Fig. 1A). Brainstem functions were evaluated by examining the pupillary state, and its responsiveness to light, during the different phases of the asphyxial and resuscitation procedures (Fig. 2). A video camera, embedded into a stereomicroscope (Leica Microsystem), was focused on the left eye and video records were synchronized with ECoG monitoring. The pupillary state was alternatively assessed in dark (1 min), and during illumination (2 min) using a 30 W spot light located at 10 cm from the eye (n = 15 protocols from 7 rats). In four experiments, the pupillary reflex was further tested by repeatedly shining light on the left eye.

2.2. Electrophysiology and data acquisition

ECoG activities were captured using silver electrodes ($\approx 60\text{ K}\Omega$) placed on the dura above S1 and M1, and a reference electrode apposed on a temporal muscle of the opposite side of the head. ECoG recordings were made from left S1 (7–8 mm anterior to the interaural line, 4.5–5.5 mm lateral to the midline) in isolation or from bilateral S1 together with the left M1 (10–11 mm anterior to the interaural line, 2.5 mm lateral to the midline) (Paxinos and Watson, 1986). ECoG signals were amplified with a differential AC amplifier (Model 1700, A-M Systems), bandpass filtered at 0.1 Hz–1 kHz and digitized at 3 kHz (CED 1401plus, Cambridge Electronic Design). Current-clamp intracellular recordings were performed simultaneously with ECoG monitoring using glass micropipettes filled with 2M KAc (60–70 M Ω) and amplified with a 900A amplifier (Molecular Devices) operating in bridge mode. Cortical neurons (n = 12) were located in the layers 5/6 of S1 (890–1572 μm below the cortical surface; Paxinos and Watson, 1986) and identified as pyramidal cells on the basis of their distinctive electrophysiological properties (Steriade, 2004). Extracellular DC potentials were obtained in S1 from the vicinity ($\approx 300\text{ }\mu\text{m}$) of the simultaneously intracellularly recorded cortical neurons. DC and intracellular signals were bandpass filtered (between DC and 3 kHz) and digitized at 10 kHz.

2.3. Asphyxial-resuscitation procedures and states segmentation

Physiological parameters and electrophysiological activities were continuously recorded during normoxic periods (control), during interruption of oxygen supply and throughout the resuscitation process. Asphyxia was induced by a sudden interruption of artificial ventilation in the curarized animals (Vent. Off). This led to a rapid collapse of EtCO₂ concentration and a more progressive fall down of

SpO₂ values towards undetectable values (< 20 % or < 10 mm Hg of oxygen pressure, corresponding to the threshold sensibility of the LifeWindow veterinary monitor) (Fig. 1A). After anoxia onset, electrical brain activities experienced transitional changes towards a sustained electrically-silent state, defined as the isoelectric period (Iso). Ventilation was then restored to resuscitate the animals (Vent. On). This procedure initiated a progressive regaining of ECoG and neuronal activities. We defined the start of the recovery period (“early recovery”) as the time point at which the ECoG signal had recovered 95% of its full amplitude, regardless of its frequency content. Subsequent recording periods were segmented according to their relative timings to the return of oxygen supply (+15 min, +30 min, +45 min, +60 min after Vent. On). Some of the animals (n = 12) were subjected to multiple asphyxial protocols, each protocol being separated from each other by a period of ECoG recovery.

2.4. Data analysis

Average membrane potential (Vm) values were calculated as the mean of the distribution of subthreshold intracellular activity and the amplitude of Vm fluctuations corresponded to the standard deviation of the distribution (SD Vm). Apparent membrane resistance (Rm) was measured from voltage responses to injection of negative current pulses (-0.5—-1 nA, 100 ms duration) applied at 5 Hz and the membrane time constant (τ_m) was derived from an exponential decay fit applied to the current-induced hyperpolarizations (Schramm et al., 2014). The mean spontaneous firing rate and mean Vm of cortical neurons were extracted from representative recording periods of 30–60 s and the temporal variability of AP discharge was assessed by calculating the inter-spike intervals coefficient of variation (CV2 ISIs), which is less dependent on variations in average firing rates than the standard CV (Altwegg-Boussac et al., 2014).

Frequency content of ECoG signals was computed during the different brain states by performing time-frequency (TF) analysis using a wavelet decomposition analysis made with custom-written functions in Matlab (The MathWorks). An ECoG signal $x(t)$ was convolved with a complex Morlet’s wavelet function defined as $w(t, f_0) = Ae^{-t^2/2\sigma_t^2} \times e^{-i2\pi f_0 t}$. Wavelets were normalized and thus $A = (\sigma_t \sqrt{\pi})^{-1/2}$. The width of each wavelet function $m = f_0/\sigma_f$ was chosen to be 7; where $\sigma_f = 1/2\pi\sigma_t$. TF contents were represented as the energy of the convolved signal: $E(t, f_0) = |w(t, f_0) \otimes x(t)|^2$, where $\otimes$ denotes the linear convolution operator. To quantify the degree of statistical association and the time delay between ECoG signals from different cortical regions, the nonlinear correlation coefficient h^2 was calculated as a function of a time shift (τ) (Lopes da Silva et al., 1989). This index quantifies the dependency of a signal $y(t)$ on a signal $x(t - \tau)$. A piecewise linear regression curve is used to predict the signal $y(t)$ given the values of $x(t - \tau)$. The variance of

$y(t)$ explained by the regression curve is called the explained variance whereas the difference with the total variance is called the unexplained variance. Thus, the nonlinear correlation ratio η^2 quantifies the reduction of variance obtained by predicting the y values on the basis of the regression curve: $\eta^2 = (\text{total variance} - \text{unexplained variance}) / \text{total variance}$. A practical implementation of this index is obtained by splitting, in a scatter plot, the values of $x(t - \tau)$ in L bins. For each bin, the average of the corresponding $y(t)$ values is calculated, the q_k points. The x value of the midpoint of each bin is called p_k . The nonlinear regression curve is then obtained by connecting the points (p_k, q_k) by segments of straight lines. The estimator of the nonlinear correlation index is computed as follows:

$$h^2(x \rightarrow y) = \frac{\sum_{i=1}^N (y_i - \langle y \rangle)^2 - \sum_{i=1}^N (y_i - g(x_i))^2}{\sum_{i=1}^N (y_i - \langle y \rangle)^2}$$

where $g(x_i)$ denotes the regression curve and $\langle y \rangle$ the average of the time series y_i over N samples. The same procedure may be carried out by splitting the values of y into bins and computing the expected values of x . The nonlinear coefficient can take values between 0 (when X and Y are independent) and 1 (in case of a perfect dependency). If the underlying relationship is linear, the coefficient h^2 approximates the common linear correlation coefficient r^2 . In contrast with a linear correlation function, this procedure yields an asymmetric function, i.e. $h^2(x \rightarrow y) \neq h^2(y \rightarrow x)$. As ECoG activities are not stationary, correlation indices were computed for short sliding epochs of 5 s with 80% of temporal overlap.

2.5. Statistics

Numerical values are given as mean $\pm$ S.E.M. Differences between groups were assessed using Kruskal-Wallis ANOVA on ranks, paired and unpaired Student's t -tests, or the non-parametric Wilcoxon signed rank test and Mann-Whitney rank sum test when appropriate (SigmaStat version 3.5, Systat Software Inc.). The significance level was set at $p < 0.05$.

3. Results

3.1. Time course of ECoG changes from asphyxia onset to cortical activity recovery

ECoG activity during control periods was dominated by rhythmic waves at 4–7 Hz (Figs. 1A, 1B and 3A), as classically observed under pentobarbital anesthesia (Altwegg-Boussac et al., 2014, 2016; Mahon and Charpier, 2012). The interruption of artificial ventilation (Vent. Off) typically resulted in a fast decline of EtCO₂, whereas the SpO₂ progressively attenuated until reaching undetectable values (see Materials and methods) within 20 s (Fig. 1A). ECoG signals remained unaffected for ≈10 s before undergoing a progressive decrease in amplitude, often preceded by a transient amplification (n = 13 out of 20 protocols, asterisk in Fig. 1B). Changes in the magnitude of the ECoG were accompanied by modifications in its frequency content. We analyzed the temporal evolution of four standard frequency bands (Buzsáki and Draguhn, 2004; Fransen et al., 2016): delta (0.5–4.5 Hz), theta (4.5–10 Hz), alpha-medium beta (10–25 Hz) and high beta-gamma (25–50 Hz). In 65% of the experiments, we identified a short lasting (9 ± 1 s, from 2 to 15 s, n = 13 protocols) surge of activity in the 10–25 and 25–50 Hz bands within 10–35 s after interruption of the ventilation, corresponding to a mean power increase relative to the pre-anoxic period of $39 \pm 10\%$ and $85 \pm 19\%$, respectively ($p < 0.01$ for both comparisons) (Fig. 1B). All frequency bands then declined towards isoelectric levels except the delta frequency band, which displayed a late transient augmentation culminating at 28–59 s (42 ± 1.7 s, n = 20 protocols) after removal of oxygen supply. This was evidenced by a marked increase (22-fold larger on average, n = 20 protocols, $p < 0.001$) in the low (0.5–4.5 Hz) to high frequency (merged 10–50 Hz) power ratio compared to control values (Fig. 1B and C, bold line).

The isoelectric state, achieved 49 ± 8 s (from 37–64 s, n = 20 protocols) after asphyxia onset, was characterized by a 93–98% decrease in ECoG amplitude associated with an almost complete suppression of cortical power in all frequency bands (Fig. 1A and D, Iso). This coincided with a flat membrane potential (Vm) in the simultaneously recorded neurons (see Fig. 4A and Supplementary Fig. 1A1), providing a direct demonstration of an electro-cerebral silence (Altwegg-Boussac et al., 2014, 2016, 2017). Notably, we observed that a large amplitude triphasic wave, similar to the previously described WoD (van Rijn et al., 2011), systematically emerged from the flat ECoG (red cross in Figs. 1A, 2A, 3A and 4A). The isoelectric state correlated with a drastic reduction of the heart rate, undetectable SpO₂ values (Fig. 1A) and a dilation of the animal's pupil, which reached a maximal areactive mydriasis at the time of the WoD (Fig. 2A and B).

Restoration of artificial ventilation (Vent. On), 2–3.5 min after asphyxia onset, led to a rapid raise of EtCO₂ and a slightly delayed (within 30–40 s) regaining of SpO₂ values (Fig. 1A). The return to

normoxic physiological parameters correlated with the appearance of another polyphasic wave of relatively small amplitude in the ECoG (inverted red cross in Figs. 1A, 2A, 3A and 4A), which inaugurated the regaining of cortical activity and was thus called the “wave-of-resuscitation” (WoR). The early phase of ECoG signal recovery, defined as the restoration of 95% of its full amplitude, was on average achieved ≈ 11 min after brain re-oxygenation and characterized by an increase in delta activity and a marked attenuation of the highest (10–50 Hz) frequency components (Figs. 1D and 3B). High-frequency oscillations then progressively resumed, approximating control values about 60 min after restoration of the ventilation (Fig. 1D). Theta activity (4.5–10 Hz) remained relatively stable, and similar to its pre-anoxic level, throughout the resuscitating process, except for a slight decrease in the corresponding ECoG power 45 min after re-oxygenation (Fig. 1D). Resuming oxygen supply also led to a progressive constriction of the dilated pupil, which returned to normal size during the early phase of ECoG recovery (Fig. 2A and B). By contrast, a lack of respiratory rescue irremediably led to the death of the animals, as attested by a continuously flat ECoG and a cardiac arrest within 6.6–15.4 min after anoxia onset ($n = 9$ experiments).

We next quantified the cortical dynamics during resuscitation by analyzing over time the coupling properties between distant cortical areas in experiments where ECoGs were simultaneously recorded from bilateral S1 and left M1 ($n = 17$ protocols) (Fig. 3C and D). The coupling index (h^2 , see Materials and methods) values between S1 and M1 on the same hemisphere were higher than those measured from bilateral S1 ($p < 0.001$) and remained similar to control values throughout the recovery period (Fig. 3D, left). The interaction strength between bilateral S1 ECoG signals was significantly increased at early recovery time and recovered pre-anoxic values 15 min after resuscitation onset (Fig. 3D, left). The analysis of time lags estimated from h^2 suggested a bidirectional coupling between bilateral S1 during all computed periods, whereas interactions between left S1 and M1 were characterized by a positive delay of about 1 ms (Fig. 3D, right).

3.2. Intracellular dynamics from asphyxia to cortical recovery

To elucidate the neuronal and synaptic mechanisms subtending the changes in ECoG activity throughout the anoxic and resuscitating periods, we made in a subset of experiments ($n = 12$) conjoint recordings of left S1 ECoG, intracellular activity of subjacent pyramidal neurons and extracellular DC potential from the vicinity of the intracellular electrode (Fig. 4A). Pyramidal cells, electrophysiologically-identified as regular-spiking ($n = 9$) or intrinsic bursting ($n = 3$) neurons (Steriade, 2004), displayed in control a sustained background synaptic activity resulting in a mean V_m of -69.6 ± 1.1 mV ($n = 12$ neurons), and a relatively low (3.1 ± 1.3 Hz, $n = 11$ active neurons) and irregular (CV2 ISIs = 1.0 ± 0.1 , $n = 11$ active neurons) firing activity (Fig. 4A–C). The surge of high-frequency ECoG

activity that immediately followed the interruption of oxygen supply was associated in pyramidal neurons with ample rhythmic **depolarizing potentials**, leading to an elevated and more regular firing compared to control periods (Fig. 4A and Supplementary Fig. 1A–C). These large **V_m fluctuations** were individually composed of small amplitude high-frequency (from 100 to 140 Hz) events, **presumably of synaptic origin**, which correlated with the fast oscillations in the corresponding ECoG (Supplementary Fig. 1C). During the subsequent low-frequency ECoG pattern, cortical neurons displayed slow composite **membrane depolarizations** that progressively attenuated in amplitude and frequency, causing a marked decrease in spontaneous firing (Fig. 4A and Supplementary Fig. 1A2, right). This vanishing of intracellular activity ended into an isoelectric V_m, deprived of any synaptic and firing events (Fig. 4A and Supplementary Fig. 1A1). The associated extracellular DC potential displayed a slow positive shift (Fig. 4A, oblique arrow), known to mirror an early anoxia-dependent increase in extracellular K⁺ (Hansen, 1985; Xie et al., 1995). The correction of V_m with extracellular DC potential values indicated an enhancement of membrane polarization by 1–6 mV (Fig. 4A, grey line).

During the isoelectric state (58–114 s after ventilation arrest), cortical neurons exhibited a sudden sharp depolarization (2.4 ± 0.3 mV/s, $n = 12$ neurons) (Fig. 4A, oblique dashed line) that coincided with an abrupt negative deflection in the adjacent DC potential and the manifestation of the WoD in the ECoG (Figs. 4A and 6A1). Membrane potential shift resulted in an anoxic depolarization in pyramidal neurons (Figs. 4A and 6A1), reaching a plateau value of -22.0 ± 3.5 mV without applying the DC correction and -3.4 ± 3.4 mV ($n = 12$ neurons) after correction (grey traces in Figs. 4A and 6A1), demonstrating an almost complete loss of transmembrane ion gradients. The neuronal anoxic depolarization, the associated isoelectric ECoG and the negative DC potential shift were maintained until the restoration of oxygen supply (Fig. 4A). This provides the first *in vivo* description of the intracellular counterpart of the anoxia-induced DC shift (Kovac et al., 2018) in the neocortex.

Resuming ventilation resulted in a slow repolarization of cortical neurons (1–38 s after Vent. On), which systematically correlated in time with the occurrence of the WoR (Figs. 4A and 6B1). After dropping to a maximal level of membrane polarization, cortical neurons were engaged in a recovery of their activity characterized by a progressive intensification of the synaptic drive that paralleled the return of ECoG waves (Fig. 4A and B). The return of background synaptic activity in cortical neurons was much slower ($p < 0.01$) than its loss and a few minutes (3.3–6.1 min) were necessary to retrieve pre-anoxic values of V_m fluctuations after restoration of oxygen supply (Fig. 4A, arrowheads and D). The early phase of ECoG recovery was associated with a change in the firing dynamics of cortical neurons, which exhibited a significantly higher spike rate and a more bursty firing profile (CV2 values > 1 in each cell; Wilbur and Rinzel, 1983), driven by rhythmic synaptic depolarizations in phase with the augmented low-frequency waves in the ECoG (Fig. 4B and C). Fifteen minutes after restoration of oxygen supply, cortical cells experienced a decrease in their ongoing firing rate towards pre-anoxic

levels and a restoration of their regular-spiking pattern (Fig. 4B). Finally, in the two neurons that could be intracellularly recorded for duration > 15 min, we found that V_m , SD V_m and spontaneous firing rate fully recovered values similar to those measured in control (data not shown).

3.3. Integrative properties of cortical neurons during anoxia and recovery process

We next investigated how the integrative properties of neocortical neurons, i.e. their ability to process membrane currents, were affected in the course of the dying-to-recovery process. We applied continuous series of negative current pulses ($n = 5$ neurons) (Fig. 5A) and measured the neuronal membrane input resistance (R_m) and time constant (τ_m), which respectively govern the ability of synaptic currents to displace V_m and the efficacy with which a cell summates synaptic events in time. As exemplified in Fig. 5A, both cellular parameters started to decline in parallel after interruption of the ventilation, falling to 61.5% (R_m) and 61.4% (τ_m) of control values at the establishment of the isoelectric V_m . R_m and τ_m values then continued to decrease and reached a maximal attenuation at the apogee of the anoxic depolarization (Fig. 5A–C). The progressive neuronal repolarization that followed the restoration of oxygen supply coincided with incremental values of R_m and τ_m , which fully regained their pre-anoxic levels when synaptic activity started to recover (Fig. 5A–C). Interestingly, we found that post-anodal depolarizing response and firing could be generated at the termination of negative current pulses during the latest part of the anoxic depolarization and in the course of membrane repolarization (Fig. 5A and D). This indicates that the regaining, during the early phases of resuscitation, of neocortical neurons intrinsic excitability could precede by more than 1 min the reappearance of synaptic and ECoG activities.

3.4. Spatiotemporal properties and intracortical correlates of the WoD and the WoR

We further examined the spatiotemporal properties and cellular correlates of the two major ECoG waveforms – the WoD and the WoR – we identified as respective cortical signatures of the beginning of, and recovery from, anoxic depolarization. By means of multi-site ECoG recordings, we found that the WoD could be recorded at spatially separated sites in the same hemisphere (left S1 and M1) as well as from homologous areas in both hemispheres (bilateral S1) (Figs. 3C and 6A1). The averages of the WoDs recorded at each cortical site demonstrated a very consistent shape, consisting of a large amplitude triphasic, initially positive, wave (Fig. 6A2). Quantification performed on the whole set of protocols indicated that the WoD latency did not significantly differ between the different cortices (from 70 to 164 s after asphyxia onset, $n = 32$ anoxia protocols, $p = 0.2$) (Fig. 6A3). However, when considering only simultaneous recordings of the left S1 and M1 ECoGs, we found that in $\approx 76\%$ of the

cases the WoD in S1 preceded the WoD in M1 by 10.1 ± 1.6 s ($n = 17$ anoxia protocols) (Fig. 6A1). As illustrated in the lower part of Fig. 6A1, showing simultaneous ECoG, *DC* potential and intracellular recordings from the left S1, the first positive deflection of the WoD was tightly correlated with the onset of both the negative shift in *DC* potential ($n = 16$ anoxia protocols) and the plateau depolarization in neocortical neurons ($n = 10$ anoxia protocols). Yet, the early phase of the intracellular anoxic depolarization, measured at 20% of its full amplitude (red dot in Fig. 6A1, lowest trace), was found to precede the corresponding ECoG WoD by 8–40 s (-26.6 ± 3.7 s, $n = 10$ anoxia protocols, $p < 0.01$), indicating that a substantial proportion of neocortical neurons were already engaged in the anoxic insult many seconds before it could be detected as a WoD in the ECoG (Fig. 6A1 and A3).

We unequivocally identified the WoR in $\approx 61\%$ (16 out of 26 resuscitation protocols) of our attempts to restore brain activity after anoxia. The presence of the WoR was systematically followed by a recovery of ECoG activity and the survival of the animal. By contrast, a lack of WoR during an unsuccessful resuscitation procedure ($n = 10$) correlated with the animal's death in all cases (see Supplementary Fig. 2). The WoR had a less regular shape and a smaller amplitude than the WoD (Fig. 6A2 versus 6B2), but its average waveform computed from the different recorded areas also revealed a consistent polyphasic profile (Fig. 6B1 and B2). As illustrated in Fig. 6B1, small amplitude electrical noise was often superimposed on ECoG records at the time of the WoR. This residual artefactual signal was not caused by neuronal activity since the simultaneously recorded cells did not display any synaptic or firing events at that time (Fig. 4A and 6B1). As for the WoD, the mean delay of occurrence of the WoR after restoration of the ventilation was similar in the three cortical areas (Fig. 6B1 and B3). The physiological relevance of the WoR as a marker of neuronal recovery was attested by its robust temporal links with the abrupt positive shift in *DC* potential and the repolarization of cortical cells at the release of the anoxic plateau potential (Fig. 6B1, lower traces, blue dots). Compared to the WoD correlates, the restoration of the *DC* potential and the repolarization of cortical cells after resuming oxygen supply exhibited more variable profiles and slower kinetics (Figs. 4A, 5A and 6B2), possibly explaining the smaller amplitude of the WoR. The onset of positive *DC* shift and cell repolarization (measured at 20% of the waveforms amplitude) was found to lag the corresponding ECoG WoR ($+1$ – 38 s for *DC* potential, $n = 6$ resuscitation protocols, $p = 0.13$; $+6$ – 42 s for *V_m*, $n = 6$ resuscitation protocols, $p < 0.05$) (Fig. 6B1 and B3). Thus, the WoR, which consistently preceded the restoration of pre-anoxic values of cell *V_m* and *DC* potential, could provide a faithful neurophysiological measure of an ongoing successful recovery of brain activity.

4. Discussion

4.1. Neuronal mechanisms from anoxia to resuscitation

In line with findings in human patients (Clute and Levy, 1990; Chawla et al., 2009; Pana et al., 2016) and in previous *in vivo* models of brain anoxia (Hansen, 1978; Vanderwolf et al., 1988; Silver and Erecinska, 1990; van Rijn et al., 2011; Borjigin et al., 2013; Kongara et al., 2014; Lee et al., 2017), we found that the interruption of oxygen supply led to a rapid (within 40–60 s) dissolution of ECoG signals, which was preceded by successive surges of fast and slow activities. The flattening of cortical signal was correlated with a decrease in the amplitude and frequency of synaptic potentials in the simultaneously recorded pyramidal neurons, consistent with the reduction of excitatory and inhibitory synaptic currents observed in neocortical cells after anoxia (Rosen and Morris, 1993; Peña and Ramirez, 2005). This dampening of synaptic activity likely resulted from the multiple anoxia-related presynaptic defects, including a reduction of presynaptic Ca^{2+} currents (Young and Somjen, 1992). The subsequent complete abolition of synaptic activities and action potential firing in cortical neurons, including interneurons since no inhibitory synaptic potentials could be detected in pyramidal cells, resulted in a flat-line ECoG (Kroeger and Amzica, 2007; Altwegg-Boussac et al., 2014, 2017). However, this suppression of synaptic events did not cause, as theoretically expected (Destexhe et al., 2003), an increase in postsynaptic membrane resistance. This possibly resulted from an additional increase in membrane conductance due to an anoxia-induced opening of ATP-dependent K^+ channels, as previously demonstrated *in vitro* from the same neuronal type we recorded (Fleidervish et al., 2001). This increase in K^+ currents may also participate to the net cell hyperpolarization observed at the establishment of the isoelectric state after subtraction of the extracellular *DC* potential shift.

The sharp membrane depolarization that inaugurated the so-called “anoxic depolarization” was presumably due to an acute effect of oxygen deprivation on cortical neurons. The general assumption is that the rapid depletion of ATP causes a failure of membrane ATP-dependent Na^+/K^+ pumps, producing a massive influx of Na^+ and efflux of K^+ , and the correlative membrane depolarization as the membrane ion gradients are lost (Hansen, 1985; Haddad and Jiang, 1993; Martin et al., 1994; Peña and Ramirez, 2005; Shimoda and Polak, 2011). The resulting depolarizing plateau potential is, in turn, likely responsible for an opening of voltage-gated Ca^{2+} channels leading to an increase in intracellular Ca^{2+} concentration, and subsequent activation of Ca^{2+} -dependent K^+ channels, further increasing the membrane conductance (Silver and Erecinska, 1990, 1992). This canonic succession of cellular processes is supported by our results showing a strong parallelism between the anoxic depolarization and the shift in the local *DC* potential, which is known to closely match the increase in extracellular K^+ and decrease in extracellular Na^+ and Ca^{2+} (Hansen 1985, Xie et al., 1995). This is also

consistent with the progressive enhancement, throughout the successive phases of the anoxic depolarization, of neocortical neurons membrane conductance, as observed in the hippocampus during brain ischemia (Xu and Pulsinelli, 1996). Unlike previous *in vitro* investigations from hippocampal cells (Tanaka et al., 1997, Dzhalal et al., 2001), synaptic events or action potentials were not systematically present during the rising phase of the anoxic depolarization. The various voltage dynamics of pyramidal neurons associated with the large depolarizing shifts could result from differences in the relative evolution of Na^+ and K^+ concentration gradients following anoxia, as recently suggested by a computer modeling study (Zandt et al., 2013).

The anoxic depolarization was maintained until artificial respiration was restored and omission of respiratory rescue irremediably led to animal death, demonstrating that our asphyxia protocol engendered a progressively developing dying process in absence of active re-oxygenation. Restoration of the ventilation initiated a recovery of neuronal membrane potential (by resupplying the ion pumps with ATP), which consisted in a gradual repolarization with highly variable time-course from cell-to-cell. This was accompanied by a retrieval of R_m and neuronal intrinsic excitability, as attested by the current-induced post-anodal firing probably due to the recovery from inactivation of transient Na^+ channels (Kuo and Bean, 1994). The subsequent slow recovery of synaptic drive and spontaneous firing closely matched the progressive regaining of ECoG activity. This indicates that the duration (from 2 to 3.5 min) of the anoxic period investigated in the present study did not preclude – without prejudging of possible post-anoxic brain damages (Rossetti et al., 2016) – the restoration of cellular and synaptic functions, [which start to recover pre-anoxic levels \$\approx\$ 60 min after reinstatement of oxygen supply.](#)

4.2. Origin of the WoD and the WoR and their clinical significance

[Asphyxia protocols were here conducted in rodents anesthetized with moderate doses of barbiturate, a pharmacological condition which could have impacted the results. However, the WoD pattern we described shares common features – including its triphasic shape, duration and latency of occurrence from the onset of brain anoxia – with that reported in awake rats after decapitation \(van Rijn et al., 2011\), thus excluding an anesthesia-dependent phenomenon.](#) It is also analogous to the terminal spreading depolarization recorded in moribund patients undergoing a global cerebral ischemia following the withdrawal of life-sustaining therapies (Dreier et al., 2018). Our intracellular recordings demonstrated its consistent temporal relationship with the neuronal anoxic depolarization and the correlative local negative *DC* potential shift, both known to reflect the anoxia-induced loss of transmembrane ions gradients (Hansen 1978, Xie et al., 1995). Assuming that a substantial number of cortical cells undergo anoxic depolarization in a nearly synchronous manner, as attested by the corresponding large negative *DC* potential shift, the raw ECoG signal is likely to approximate the

summed contributions of individual neurons and the triphasic shape of the WoD to originate from the high-pass filtering of the resulting potential with a 0.1-Hz cut-off frequency (van Rijn et al., 2011; Zandt et al., 2011; Dreier et al., 2017). The collective repolarization of cortical neurons associated with a positive shift in the DC potential after restoration of oxygen supply, likely due to the restoration of ion gradients, was reflected as a second polyphasic wave in the ECoG, the WoR. The relatively low amplitude of the WoR was probably due to the high cell-to-cell variability in delay and time course of Vm recovery, resulting in smaller current flows within the cortical tissue. In most of experiments where both S1 and M1 ECoGs were simultaneously recorded, the WoD first appeared in S1. This could result from the high vulnerability to hypoxia-ischemia of this cortical region (Martin et al., 1997), which may then propagate the anoxic depolarization wave, *via* spreading depression-like processes (Bogdanov et al., 2016), to distant cortical areas.

Restoration of oxygen supply, after the occurrence of the WoD, was followed in most cases (~61%) by a WoR that systematically inaugurated the recovery of neuronal and ECoG activity. This suggests that the WoD cannot be considered as a biomarker of a massive neuronal death or irremediable brain death, and thus may be more appropriately renamed “wave of anoxic depolarization”. However, we can conclude from the present study that the WoD provides a reliable signature, in both rodents and humans, of a severe anoxic insult of neocortical neurons that can be readily recorded from the brain surface. To our knowledge, the present work is the first to describe the existence and the cellular correlates of a surface cortical wave predictive of a post-anoxic cortical recovery: the WoR. This wave represents a faithful anticipatory biomarker of successful resuscitation, which can precede by ~20 s the onset of Vm recovery in individual cortical neurons. By means of appropriate EEG filtering in human patients undergoing resuscitation procedures, the identification of a WoR, or a lack of detection, may potentially participate to the prognosis of a good post-anoxic outcome or to the diagnosis of an irreversible brain deterioration.

4.3. A new neuronal basis for near-death experiences?

Our protocols of asphyxia rapidly led to the establishment of an isoelectric state, shortly preceded by successive surges fast and slow oscillations in the ECoG. Such transient increases in brain electrical activity at near-death have been previously observed in non-anaesthetized rodents after sudden brain anoxia (Vanderwolf et al., 1988; Borjigin et al., 2013) and suggested by bispectral index measurements after withdrawal of life support in human patients (Chawla et al., 2009, 2017; Auyong et al., 2010). The increase in high-frequency oscillations was interpreted as a possible correlate of a heightened conscious state that could account for NDEs (Chawla et al., 2009, 2017; Borjigin et al., 2013). Our findings challenge this hypothesis. We found that the surge of beta-gamma ECoG activities correlated

in pyramidal neurons with regular firing at 6-7 Hz driven by rhythmic membrane oscillations, which were shaped by the summation of high-frequency (up to 140 Hz) depolarizing and hyperpolarizing synaptic potentials. This pattern is similar to the transient burst of fast activity previously recorded *in vivo* in hippocampal networks during cerebral ischemia (Freund et al., 1989) and shown to result from an oscillatory synaptic excitation tuned by a GABA- and adenosine-dependent inhibition (Dzhala et al., 2001). Regardless of a possible effect of anesthesia, which however does not preclude the occurrence of NDEs (Lopez et al., 2006), the neuronal and synaptic activities we recorded in association with the increase in high-frequency ECoG oscillations differed from that of neocortical neurons during waking state, ordinarily characterized by a sustained depolarizing synaptic drive and an irregular firing (Steriade et al., 2001). Moreover, the duration of the high-frequency surge (a few seconds), as estimated by the bispectral index score in humans, is seemingly too short to explain the richness and the temporal extent of the narratives from patients who experienced NDE (Owens et al., 1990; van Lommel et al., 2001). It is also improbable that NDEs occur from the transient isoelectric brain state – and associated WoD – since we demonstrated that cortical neurons had lost their excitability and were not spontaneously active during this state, and therefore, unable to generate conscious experience.

It is conspicuous that NDE reports require successful resuscitation and recovery of cortical and cognitive functions. We found that the time necessary to recover normoxic synaptic and firing patterns in cortical neurons and networks was much longer than that corresponding to anoxia-induced activity decline, making this period more suitable to subtend NDEs. The slow process of recovery was characterized by overwhelming 5–10 Hz oscillations in the ECoG and intracellular activities, together with persisting attenuated high-frequency patterns. Comparable atypical patterns of brain oscillations, associating a decrease in gamma activities with an increase in low-frequency pattern, have been reported in hallucinatory diseases, such as schizophrenia (Uhlhaas and Singer 2010). It is thus plausible that the long-lasting disorganization of cortical dynamics characterized by a deficit in fast ECoG activity, especially in sensory areas, be responsible for the stereotyped, hallucinatory-like (Morse et al., 1989), mental imagery that characterize some typical elements of NDEs. We thus suggest that the slow recovery of brain activities following resuscitation may represent a relevant time period to get deeper insight into the neuronal basis of NDEs.

5. Conclusions

Our multi-scale electrophysiological recordings combined with the monitoring of vital parameters demonstrate that brain anoxia triggers a temporally-organized sequence of synaptic and firing patterns in the neocortex which ends with the establishment of an isoelectric state deprived of neuronal activity. About 1 min after the silencing of cortical activities, neocortical neurons strongly depolarize

to a Vm where firing is no longer observed and cell excitability is lost. We provide evidence that this state of depolarization block does not imply irreversible neuronal death since a recovery of cell membrane polarization can be initiated if re-oxygenation is rapidly reinstated. Membrane potential repolarization upon resuscitation is accompanied by a return of cell integrative properties followed by the progressive regaining of normoxic synaptic and firing activities in neurons and networks. The anoxia-induced massive depolarization of neocortical cells and its reversal upon return of oxygen supply respectively results in large amplitude positive and negative shifts of the cortical *DC* potential, which can be captured as reproducible polyphasic ECoG waveforms at the surface of the brain: the WoD and the WoR.

Unlike the isoelectric EEG, which does not provide unequivocal information on neuronal excitability (Altwegg-Boussac et al., 2017) and on the timing of cytotoxic cellular processes after anoxia, the present findings demonstrate that the WoD and the WoR provide reliable and easily recordable biomarkers of the onset of, and recovery, from anoxia-related neuronal insult. Although an equivalent of the WoR has never been recorded from human patients so far, it is very likely that a collective recovery of Vm would also occur in human cortical neurons upon successful cardiopulmonary resuscitation. Therefore, the WoD and the WoR may help refine interventional emergency procedures as they could be used as a real-time pathophysiological marker of brain death initiating processes (WoD) and as a cortical signature for positive recovery prognosis (WoR). In consequence, and ultimately, these results may also have implications for ethical and clinical debates on organ donation involving dying patients.

6. Author contributions

A.E.S., V.N., S.M., and S.C. designed research; A.E.S. and A.C.L. performed research; A.E.S., S.D., M.C., A.C.L., and S.M. analyzed data; S.M. and S.C. supervised research and wrote the paper.

Declaration of competing Interest

The authors declare no conflict of interest.

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References

- Altwegg-Boussac, T., Chavez, M., Mahon, S., Charpier, S., 2014. Excitability and responsiveness of rat barrel cortex neurons in the presence and absence of spontaneous synaptic activity in vivo. *J. Physiol.* 592:3577–95.
- Altwegg-Boussac, T., Mahon, S., Chavez, M., Charpier, S., Schramm, A.E., 2016. Induction of an Isoelectric Brain State to Investigate the Impact of Endogenous Synaptic Activity on Neuronal Excitability In Vivo. *J. Vis. Exp.* 31:e53576.
- Altwegg-Boussac, T., Schramm, A.E., Ballesterio, J., Grosselin, F., Chavez, M., Lecas, S., Baulac, M., Naccache, L., Demeret, S., Navarro, V., Mahon, S., Charpier, S., 2017. Cortical neurons and networks are dormant but fully responsive during isoelectric brain state. *Brain* 140:2381–2398.
- Auyong, D.B., Klein, S.M., Gan, T.J., Roche, A.M., Olson, D., Habib, A.S., 2010. Processed electroencephalogram during donation after cardiac death. *Anesth. Analg.* 110:1428–1432.
- Bogdanov, V.B., Middleton, N.A., Theriot, J.J., Parker, P.D., Abdullah, O.M., Ju, Y.S., Hartings, J.A., Brennan, K.C., 2016. Susceptibility of Primary Sensory Cortex to Spreading Depolarizations. *J. Neurosci.* 36:4733–4743.
- Borjigin, J., Lee, U., Liu, T., Pal, D., Huff, S., Klarr, D., Sloboda, J., Hernandez, J., Wang, M.M., Mashour, G.A., 2013. Surge of neurophysiological coherence and connectivity in the dying brain. *Proc. Natl. Acad. Sci. U. S. A.* 110:14432–14437.
- Buzsáki, G., Draguhn, A., 2004. Neuronal oscillations in cortical networks. *Science* 304:1926–1929.
- Chawla, L.S., Akst, S., Junker, C., Jacobs, B., Seneff, M.G., 2009. Surges of electroencephalogram activity at the time of death: a case series. *J. Palliat. Med.* 12:1095–1100.
- Chawla, L.S., Terek, M., Junker, C., Akst, S., Yoon, B., Brasha-Mitchell, E., Seneff, M.G., 2017. Characterization of end-of-life electroencephalographic surges in critically ill patients. *Death Stud.* 41:385–392.
- Clute, H.L., Levy, W.J., 1990. Electroencephalographic changes during brief cardiac arrest in humans. *Anesthesiology* 73:821–825.
- Destexhe, A., Rudolph, M., Paré, D., 2003. The high-conductance state of neocortical neurons in vivo. *Nat. Rev. Neurosci.* 4:739–751.
- Dreier, J.P., Major, S., Foreman, B., Winkler, M.K.L., Kang, E.J., Milakara, D., Lemale, C.L., DiNapoli, V., Hinzman, J.M., Woitzik, J., Andaluz, N., Carlson, A., Hartings, J.A., 2018. Terminal spreading depolarization and electrical silence in death of human cerebral cortex. *Ann. Neurol.* 83:295–310.
- Dreier, J.P., Fabricius, M., Ayata, C., Sakowitz, O.W., Shuttleworth, C.W., Dohmen, C., et al., 2017. Recording, analysis, and interpretation of spreading depolarizations in neurointensive care: Review and recommendations of the COSBID research group. *J. Cereb. Blood. Flow Metab.* 37:1595–1625.
- Dzhala, V., Khalilov, I., Ben-Ari, Y., Khazipov, R., 2001. Neuronal mechanisms of the anoxia-induced network oscillations in the rat hippocampus in vitro. *J. Physiol.* 536:521–531.

- Fleidervish, I.A., Gebhardt, C., Astman, N., Gutnick, M.J., Heinemann, U., 2001. Enhanced spontaneous transmitter release is the earliest consequence of neocortical hypoxia that can explain the disruption of normal circuit function. *J. Neurosci.* 21:4600–4608.
- Fransen, A., Dimitriadis, G., van Ede, F., Maris, E., 2016. Distinct α - and β -band rhythms over rat somatosensory cortex with similar properties as in humans. *J. Neurophysiol.* 115:3030–3044.
- Freund, T.F., Buzsáki, G., Prohaska, O.J., Leon, A., Somogyi, P., 1989. Simultaneous recording of local electrical activity, partial oxygen tension and temperature in the rat hippocampus with a chamber-type microelectrode. Effects of anaesthesia, ischemia and epilepsy. *Neuroscience* 28, 539–549.
- Haddad, G.G., Jiang, C., 1993. O₂ deprivation in the central nervous system: on mechanisms of neuronal response, differential sensitivity and injury. *Prog. Neurobiol.* 40:277–318.
- Hansen, A.J., 1978. The extracellular potassium concentration in brain cortex following ischemia in hypo- and hyperglycemic rats. *Acta Physiol. Scand.* 102:324–329.
- Hansen, A.J., 1985. Effect of anoxia on ion distribution in the brain. *Physiol. Rev.* 65:101–148.
- Kaminogo, M., Suyama, K., Ichikura, A., Onizuka, M., Shibata, S., 1998. Anoxic depolarization determines ischemic brain injury. *Neurol. Res.* 20:343–348
- Kongara, K., McIlhone, A.E., Kells, N.J., Johnson, C.B., 2014. Electroencephalographic evaluation of decapitation of the anesthetized rat. *Lab. Anim.* 48:15–19.
- Kovac, S., Speckmann, E.J., Gorji, A., 2018. Uncensored EEG: The role of DC potentials in neurobiology of the brain. *Prog. Neurobiol.* 165–167:51–65.
- Kroeger, D., Amzica, F., 2007. Hypersensitivity of the anesthesia-induced comatose brain. *J. Neurosci.* 27:10597–10607.
- Kuo, C.C., Bean, B.P., 1994. Na⁺ channels must deactivate to recover from inactivation. *Neuron* 12:819–829.
- Laureys, S., 2005. Science and society: death, unconsciousness and the brain. *Nat. Rev. Neurosci.* 6:899–909.
- Lee, D.E., Lee, L.G., Siu, D., Bazrafkan, A.K., Farahabadi, M.H., Dinh, T.J., Orellana, J., Xiong W., Lopour, B.A., Akbari, Y., 2017. Neural Correlates of Consciousness at Near-Electrocerebral Silence in an Asphyxial Cardiac Arrest Model. *Brain Connect.* 7:172–181.
- Lipton, P., 1999. Ischemic cell death in brain neurons. *Physiol. Rev.* 79:1431–1568.
- Lopes da Silva, F., Pijn, J.P., Boeijinga, P., 1989. Interdependence of EEG signals - linear vs nonlinear associations and the significance of time delays and phase shifts. *Brain Topography* 2:9–18
- Lopez, U., Forster, A., Annoni, J.M., Habre, W., Iselin-Chaves, I.A., 2006. Near-death experience in a boy undergoing uneventful elective surgery under general anesthesia. *Paediatr. Anaesth.* 16, 85–88.
- Mahon, S., Charpier, S., 2012. Bidirectional plasticity of intrinsic excitability controls sensory inputs efficiency in layer 5 barrel cortex neurons in vivo. *J. Neurosci.* 32:11377–11389.

- Martin, R.L., Lloyd, H.G., Cowan, A.I., 1994. The early events of oxygen and glucose deprivation: setting the scene for neuronal death? *Trends Neurosci.* 17:251–257.
- Martin, L.J., Brambrink, A., Koehler, R.C., Traystman, R.J., 1997. Primary sensory and forebrain motor systems in the newborn brain are preferentially damaged by hypoxia-ischemia. *J. Comp. Neurol.* 377:262–285.
- Morse, M.L., Venecia, D., Jr., Milstein, J., 1989. Near-death experiences: A neurophysiologic explanatory model. *Journal of Near-Death Studies* 8, 45–53.
- Owens, J.E., Cook, E.W., Stevenson, I., 1990. Features of "near-death experience" in relation to whether or not patients were near death. *Lancet* 336, 1175–1177.
- Pana, R., Hornby, L., Shemie, S.D., Dhanani, S., Teitelbaum, J., 2016. Time to loss of brain function and activity during circulatory arrest. *J. Crit. Care* 34:77–83.
- Paxinos, G., Watson, C., 1986. *The Rat Brain in Stereotaxic Coordinates*. Academic Press, New York.
- Peña, F., Ramirez, J.M., 2005. Hypoxia-induced changes in neuronal network properties. *Mol. Neurobiol.* 32:251–283.
- Ritter, A.M., Muizelaar, J.P., Barnes, T., Choi, S., Fatouros, P., Ward, J., Bullock, M.R., 1999. Brain stem blood flow, pupillary response, and outcome in patients with severe head injuries. *Neurosurgery* 44:941–948.
- Rosen, A.S., Morris, M.E., 1993. Anoxic depression of excitatory and inhibitory postsynaptic potentials in rat neocortical slices. *J. Neurophysiol.* 69: 109–117
- Rossetti, A.O., Rabinstein, A.A., Oddo, M., 2016. Neurological prognostication of outcome in patients in coma after cardiac arrest. *Lancet Neurol.* 15:597–609.
- Schramm, A.E., Marinazzo, D., Gener, T., Graham, L., 2014. The Touch and Zap method for in vivo whole-cell patch recording of intrinsic and visual responses of cortical neurons and glial cells. *PLoS One* 9, e97310.
- Shimoda, L.A., Polak, J., 2011. Hypoxia. 4. Hypoxia and ion channel function. *Am. J. Physiol. Cell Physiol.* 300:C951–967.
- Seth, A.K., Baars, B.J., Edelman, D.B., 2005. Criteria for consciousness in humans and other mammals. *Conscious. Cogn.* 14:119–139.
- Silver, I.A., Erecińska, M., 1990. Intracellular and extracellular changes of $[Ca^{2+}]$ in hypoxia and ischemia in rat brain in vivo. *J. Gen. Physiol.* 95:837–866.
- Silver, I.A., Erecińska, M., 1992. Ion homeostasis in rat brain in vivo: intra- and extracellular $[Ca^{2+}]$ and $[H^{+}]$ in the hippocampus during recovery from short-term, transient ischemia. *J. Cereb. Blood Flow Metab.* 12:759–772.
- Steriade, M., Timofeev, I., Grenier, F., 2001. Natural waking and sleep states: a view from inside neocortical neurons. *J. Neurophysiol.* 85:1969–1685.
- Steriade, M., 2004. Neocortical cell classes are flexible entities. *Nat. Rev. Neurosci.* 5:121–34.

Tanaka, E., Yamamoto, S., Kudo, Y., Mihara, S., Higashi, H., 1997. Mechanisms Underlying the Rapid Depolarization Produced by Deprivation of Oxygen and Glucose in Rat Hippocampal CA1 Neurons In Vitro. *J. Neurophysiol.* 78: 891–902

Uhlhaas, P.J., Singer, W., 2010. Abnormal neural oscillations and synchrony in schizophrenia. *Nat. Rev. Neurosci.* 11:100–113.

Vanderwolf, C.H., Buzsaki, G., Cain, D.P., Cooley, R.K., Robertson, B., 1988. Neocortical and hippocampal electrical activity following decapitation in the rat. *Brain Res.* 451:340–344.

van Lommel, P., van Wees, R., Meyers, V., Elfferich, I., 2001. Near-death experience in survivors of cardiac arrest: a prospective study in the Netherlands. *Lancet* 358:2039–2045.

van Rijn, C.M., Krijnen, H., Menting-Hermeling, S., Coenen, A.M., 2011. Decapitation in rats: latency to unconsciousness and the 'wave of death'. *PLoS One* 6:e16514.

Wilbur, W.J., Rinzel, J., 1983. A theoretical basis for large coefficient of variation and bimodality in neuronal interspike interval distributions. *J. Theor. Biol.* 105, 345–68.

Xie, Y., Zacharias, E., Hoff, P., Tegtmeier, F., 1995. Ion channel involvement in anoxic depolarization induced by cardiac arrest in rat brain. *J. Cereb. Blood Flow Metab.* 15:587–594.

Xu, Z.C., Pulsinelli, W.A., 1996. Electrophysiological changes of CA1 pyramidal neurons following transient forebrain ischemia: an in vivo intracellular recording and staining study. *J. Neurophysiol.* 76:1689–1697.

Young, J.N., Somjen, G.G., 1992. Suppression of presynaptic calcium currents by hypoxia in hippocampal tissue slices. *Brain Res.* 573:70–76.

Zandt, B.J., ten Haken, B., van Dijk, J.G., van Putten, M.J., 2011. Neural dynamics during anoxia and the "wave of death". *PLoS One* 6:e22127.

Zandt, B.J., Stigen, T., Ten Haken, B., Netoff, T., van Putten, M.J., 2013. Single neuron dynamics during experimentally induced anoxic depolarization. *J. Neurophysiol.* 110:1469–75.

Fig. 1. Time-dependent changes in S1 ECoG activity from asphyxia onset to recovery of a pre-anoxic cortical state. **(A)** Typical ECoG recording from the left S1, together with the monitoring of oxygen saturation (SpO₂), end-tidal carbon dioxide concentration (EtCO₂), core temperature (Temp.) and heart rate, during pre-asphyxia period, oxygen deprivation and throughout the process of recovery. The interruption of the artificial ventilation (Vent. Off) and its restoration (Vent. On) are indicated by the vertical dashed lines. A time-frequency map of the ECoG signal is provided below the ECoG record. Recordings are discontinued (horizontal dashed lines) to properly illustrate the recovery of cortical activity 60 min after Vent. On. The blue box indicates the time period corresponding to a 95% recovery of the full amplitude of ECoG signal (Early recovery). Here, and in the following figures, $\uparrow$ and $\downarrow$ indicate the WoD and the WoR, respectively. **(B)** Expansion of an ECoG epoch prior to, and after, interruption of the ventilation (Vent. Off) and corresponding time-frequency map. The graph below indicates the relative power of the different frequency bands (normalized as a function of their mean control value computed over a 2-min period) over time. Note the surges of high-frequency oscillations during the early anoxic period (asterisk), followed by a bump of low frequency (0.5–4.5 Hz) activity. The bold line represents the power ratio (Γ) of low frequencies (LF, 0.5–4.5 Hz) to merged faster components (HF, 10–50 Hz). In this experiment, the peak (p.) of LF/HF ratio after anoxia onset corresponded to an augmentation of 9,800% relative to the control value. **(C)** Population data showing the time-course of the LF/HF ratios, aligned to their peak values (p.) (n = 20 protocols). The bold line indicates the mean value computed over all experiments. The graph at right depicts the LF/HF ratios for individual experiments (grey lines), in control (Ctrl.) and at the peak value during asphyxia (p.). The black line connects the corresponding mean values ($\pm$ S.E.M). **(D)** Summary data illustrating the mean ($\pm$ S.E.M) ECoG power for the different frequency bands (computed over 30 s for the isoelectric period or 1 min for the other states, and normalized to a 2-min control period) in control (Ctrl., n = 20 protocols), isoelectric state (Iso, n = 20 protocols), early recovery period (Early, n = 20 protocols), as well as +15 min (n = 16 protocols), +30 min (n = 10 protocols) and +45–60 min (n = 5 protocols) after restoration of oxygen supply. Time-dependent changes in ECoG power were normalized as a function of the corresponding control value. *p < 0.05, **p < 0.01, ***p < 0.001, n.s. non-significant. Paired Student t-tests for all comparative measurements. Records in (A) and (B) are from two different experiments.

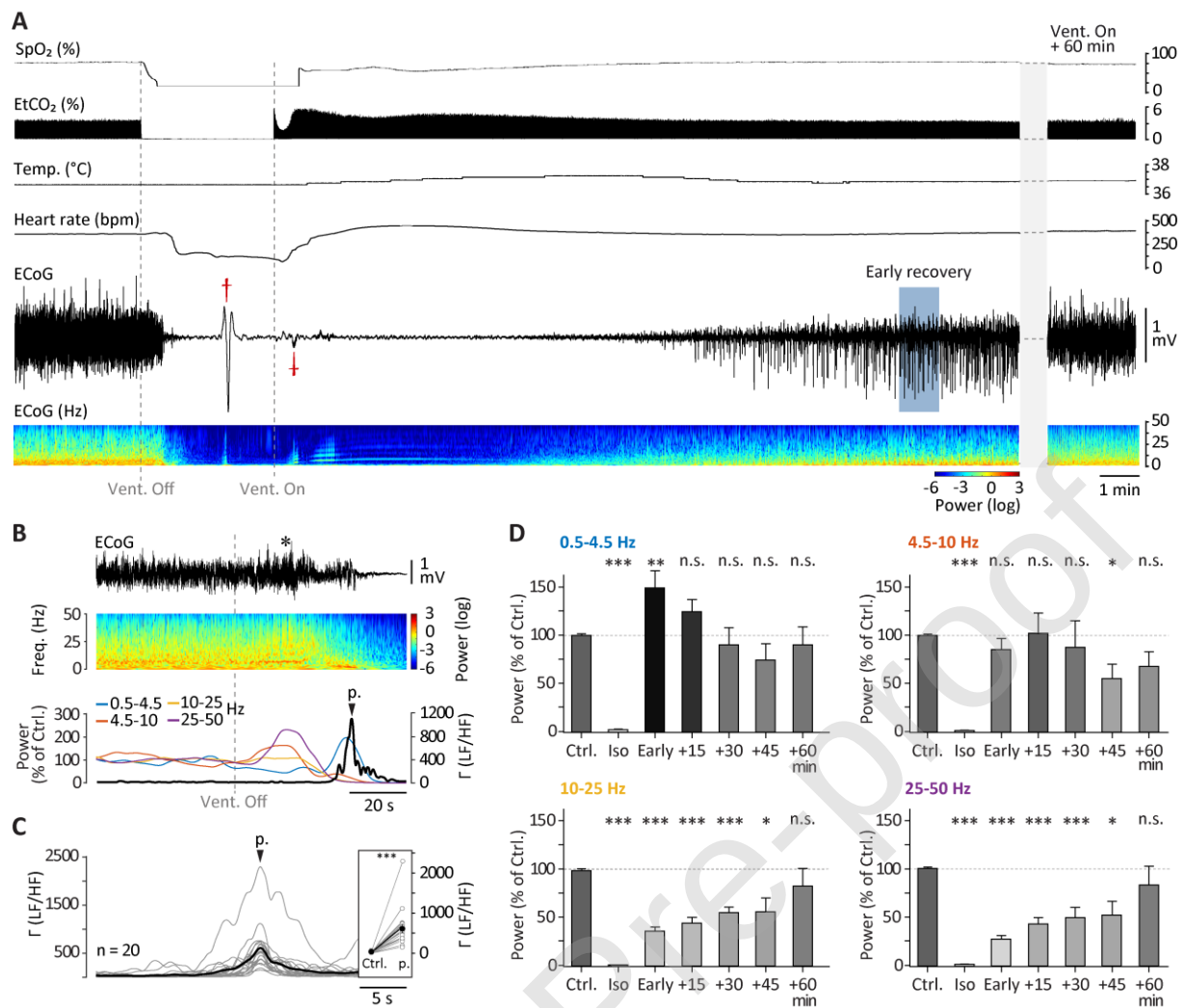


Fig. 2. Time course of changes in the pupillary state during brain anoxia and resuscitation. **(A)** S1 ECoG profile during an asphyxia protocol and the subsequent recovery process. Grey dashed lines indicate the times at which ventilation was turned off (Vent. Off) and reinstated (Vent. On). **(B)** Instantaneous video pictures taken from the illuminated left eye (see Materials and methods) at the times indicated by the arrows in (A). Note the constriction of the pupil in response to light under normoxic condition (1) and the large non-reactive mydriasis during the anoxic isoelectric ECoG (2), which reached a climax in conjunction with the WoD (3). This mydriatic state and the associated lack of pupillary reflex were suggestive of a disruption of the third cranial nerve nucleus function caused by brainstem anoxia (Ritter et al., 1999). About 1 min after the occurrence of the WoR, the pupil started to re-constrict (4) and recovered its pre-anoxic state, as well as a normoxic light reflex, at the early recovery time (5).

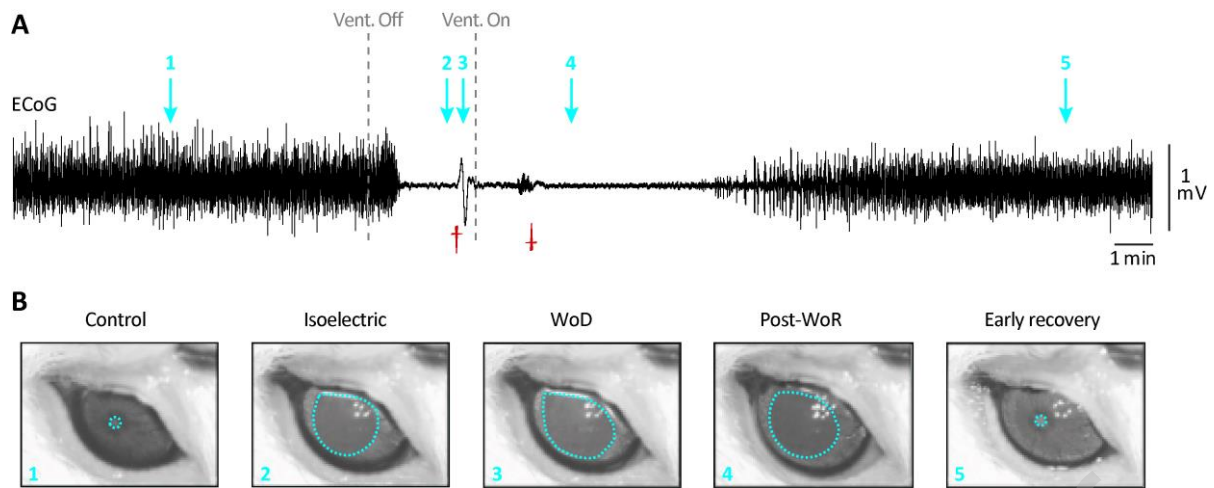


Fig. 3. Dynamics of ECoG activities and changes in functional coupling between cortical areas during resuscitation process. **(A)** Representative ECoG recording from left S1, during pre-asphyxia period (Ctrl.), oxygen deprivation and throughout the recovery process. The interruption of artificial ventilation (Vent. Off) and its restoration (Vent. On) are indicated by the grey dashed lines. **(B)** Expansion of the ECoG segments (top), as indicated by the blue boxes in (A), corresponding time-frequency maps (middle) and power of the indicated frequency bands relative to their control values. **(C)** Simultaneous recordings of end-tidal carbon dioxide concentration (EtCO₂) and multisite ECoGs from the indicated cortical regions, during control period, after interruption of the artificial ventilation and in the course of cortical recovery following re-oxygenation. **(D)** Cortico-cortical nonlinear activity coupling and associated time delays. The data are reported as mean $\pm$ S.E.M for each computed period. Measurements were made in control (Ctrl.), during the early recovery of ECoG activities and at different periods after restoration of the ventilation. Left, The degree of coupling between cortical areas was assessed by calculating the nonlinear correlation coefficient (h^2 , see Materials and methods) extracted for successive 5-s windows with a temporal overlap of 80%. The reported values are the average of $h^2(X \rightarrow Y)$ and $h^2(Y \rightarrow X)$. Right, Corresponding time delays. The time lag values suggested a bidirectional coupling between bilateral S1 during all computed periods and a unidirectional coupling between left S1 and ipsilateral M1 with a delay around ≈ 1 ms. $**p < 0.01$. Statistical significance was assessed using paired Student *t*-tests.

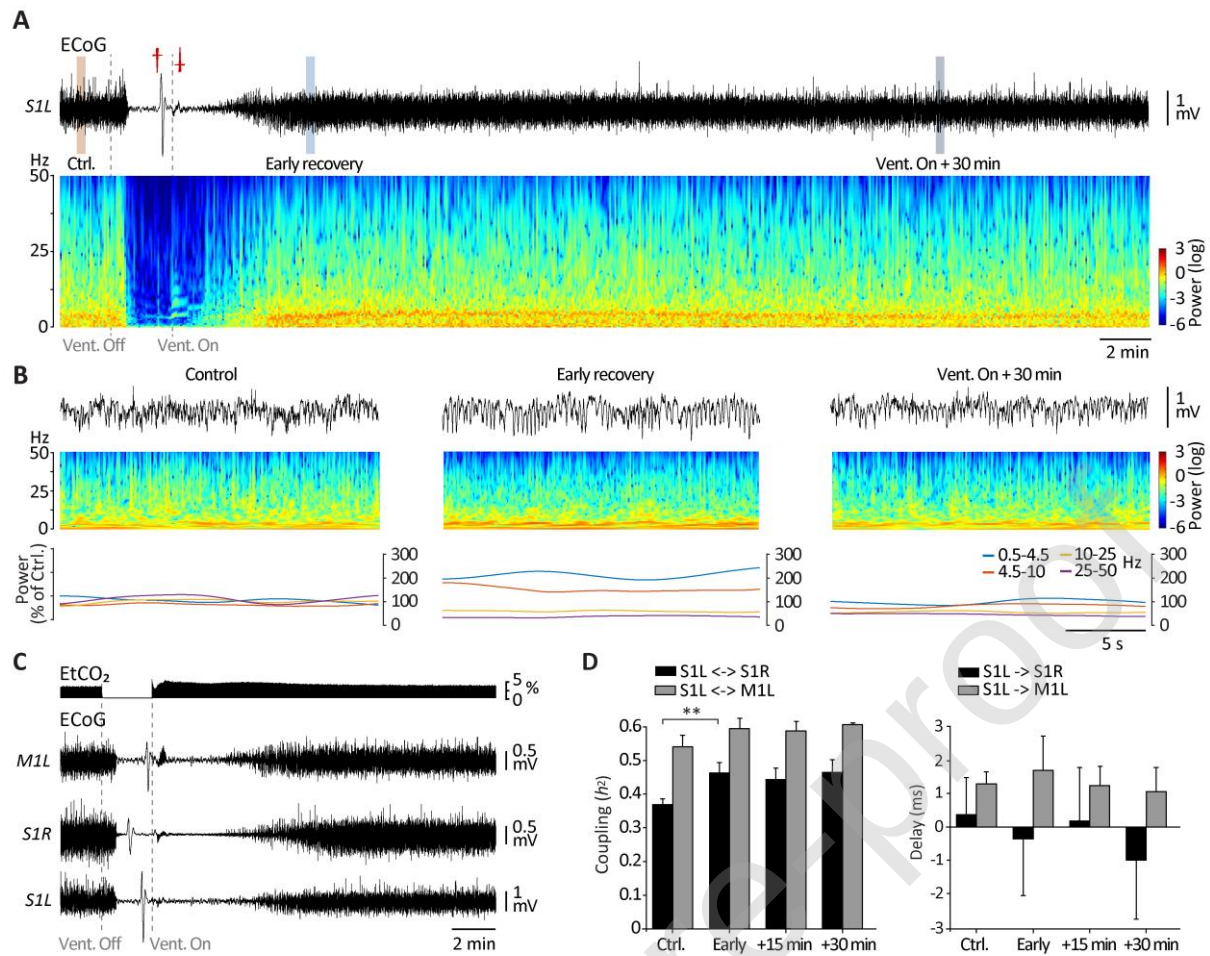


Fig. 4. Dynamic changes of neocortical neurons intracellular activity from the onset of asphyxia to the recovery of brain activity. (A) Simultaneous recordings of S1 ECoG (top), intracellular activity from a subjacent pyramidal cell (middle) and local DC potential (bottom), from control period, during oxygen deprivation, up to the recovery of cortical activity. The grey trace corresponds to the Vm corrected with the corresponding DC potential shift. The vertical dashed lines indicate the onset of asphyxia (Vent. Off) and the restoration of artificial respiration (Vent. On). The corresponding values of SD Vm are provided below the intracellular record. Recordings are discontinued to show the neuronal activity 15 min after re-oxygenation onset. The slope of the anoxic depolarizations was calculated using best linear fitting procedures (oblique orange dashed line) ($r^2 = 0.99$). **(B)** Enlargement of the ECoG (top) and associated intracellular segments (bottom) as indicated in (A), in control (1), during early recovery (2) and 15 min after restoration of oxygen supply (3). **(C)** Pooled data showing the values (mean $\pm$ S.E.M) of Vm ($n = 12$ neurons), spontaneous firing rate ($n = 11$ neurons) and corresponding regularity of discharge (CV2 ISIs, $n = 11$ neurons), in control (Ctrl.) and during the early phase of recovery (Early). Vm values were corrected according to the corresponding DC potential shift. **(D)** Summary data comparing the time lapses at the transition towards isoelectric Vm (orange arrowheads) and at the transition towards Vm recovery (blue arrowheads) (left), and the corresponding rate of Vm shifts as

function of time (dV_m/dt) (right). * $p < 0.05$, ** $p < 0.01$, n.s. non-significant. Statistical significance was assessed using unpaired Student t -test or the non-parametric Mann-Whitney rank sum test.

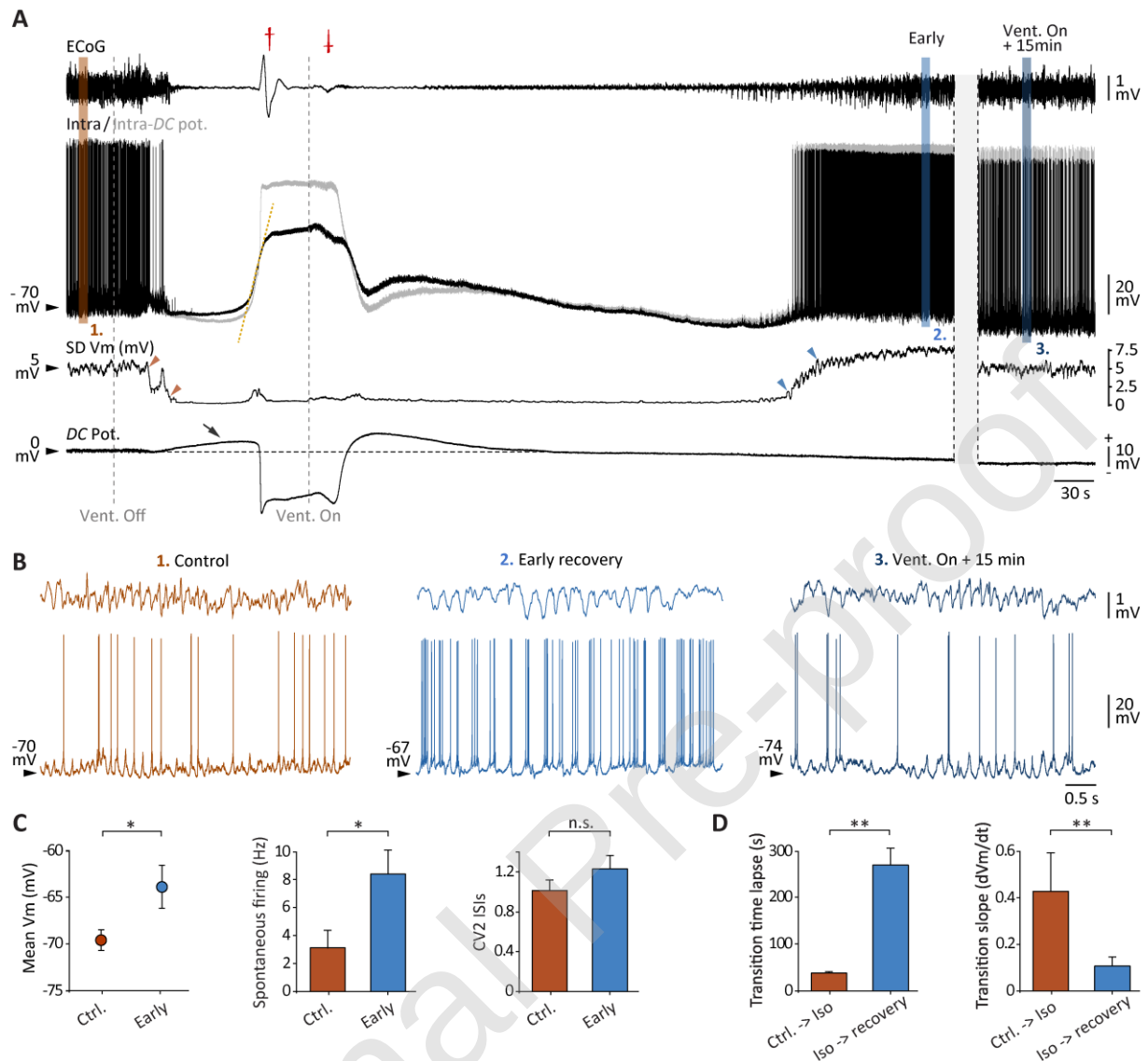


Fig. 5. Time course of changes in neuronal integrative properties. (A) Spontaneous activity and voltage responses of a S1 pyramidal neuron to iterative injections of square negative current pulses in control, during anoxia period (edged by the dashed lines) and in the course of cortical recovery. The simultaneously recorded oxygen saturation (SpO_2) and end-tidal carbon dioxide concentration ($EtCO_2$) are shown at the top. The grey trace is the V_m corrected by the adjacent DC potential (*not shown*). The data points below the intracellular record indicate the values of R_m and τ_m over time. **(B)** Mean ($n = 10$ successive trials) V_m changes ($\pm$ S.E.M) in response to negative current pulses (bottom), at the times indicated in (A): in control (1), during the isoelectric state before the anoxic depolarization (AD) (2), during the neuronal AD plateau potential (3) and at the early recovery time (4). Note the complete lack of variability in current-evoked membrane voltage deflections during the anoxic epochs (2. and 3.). **(C)** Summary data comparing the values of R_m and τ_m calculated from the periods indicated in (B). **(D)**

Examples of sub- (green) and supra-threshold (black) depolarizing responses at the termination of the hyperpolarizing current pulses, during the post-anoxic cell repolarization (crosses in (A)). The V_m values at current pulses offset are indicated. ** $p < 0.01$; *** $p < 0.001$, n.s. non-significant. Significance was assessed using paired and unpaired Student t -tests.

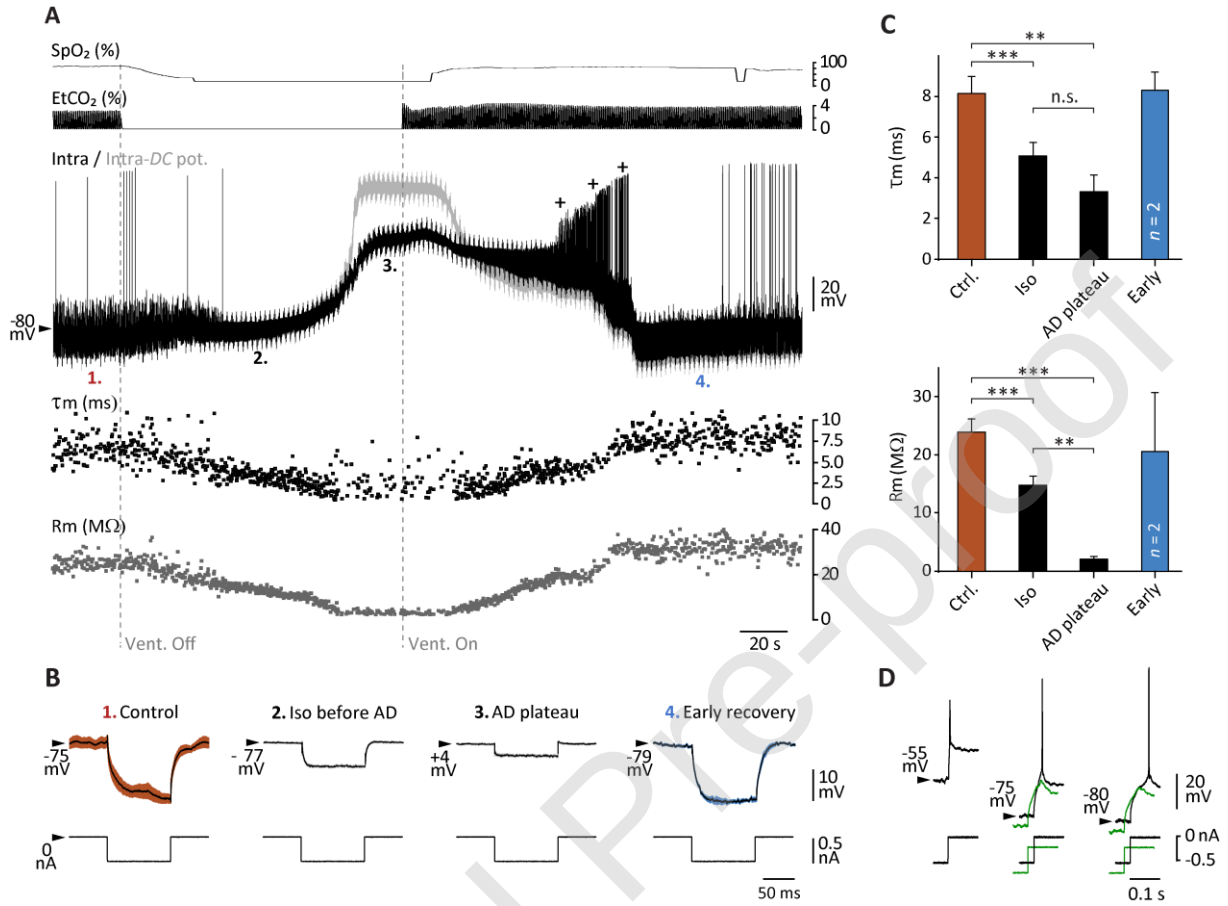


Fig. 6. Spatiotemporal properties and intracortical correlates of the WoD and the WoR. (A) WoD. (A1) Simultaneous recordings (from top to bottom) of end-tidal carbon dioxide concentration (EtCO₂), multisite ECoGs, DC potential and intracellular activity from a pyramidal neuron, during control and a 2-min period after asphyxia onset. The dashed vertical line indicates the interruption of artificial ventilation (Vent. Off). The grey trace is the V_m rectified by the corresponding DC potential shift. S1L and S1R: left and right primary somatosensory cortex, M1L: left primary motor cortex. (A2) Superimposition of individual WoDs ($n = 32$ anoxia protocols) and DC potentials ($n = 16$ anoxia protocols) recorded from the cortical areas indicated in (A1), temporally aligned on the peak of the first positive component of the WoD, or on the midpoint of DC potential negative deflection (DC pot.). The corresponding average waveforms (bold traces) are superimposed. (A3) Pooled data comparing the delays of occurrence of ECoG WoDs from the three cortical regions ($n = 32$ anoxia protocols, $p = 0.2$), S1L ECoG WoD and corresponding DC potential ($n = 16$ anoxia protocols, $p = 0.36$), and S1L ECoG WoD and neuronal anoxic depolarization ($n = 10$ anoxia protocols, $p = 0.001$) after interruption of

oxygen supply. Latencies of WoD, DC shift and cell anoxic depolarization were measured at 20% of their full amplitude (red dots in (A1)). **(B)** WoR. (B1) Same representation as in (A1). The traces correspond to the 1.5-min recording period that followed the restoration of the ventilation (Vent. On). (B2) Same representation as in (A2) for the ECoG WoRs recorded from both S1 and left M1 ($n = 16$ resuscitation protocols). (B3) Population data comparing the delays of occurrence of ECoG WoRs ($n = 16$ resuscitation protocols, $p = 0.88$) recorded from both S1 and left M1, S1L ECoG WoR and DC potential recovery ($n = 6$ resuscitation protocols, $p = 0.13$), and S1L ECoG WoD and neuronal post-anoxic repolarization ($n = 6$ resuscitation protocols, $p = 0.01$) after restoration of oxygen supply. Latencies of WoR, DC positive shift and cell post-anoxic repolarization were measured at 20% of their full amplitude (blue dots in (B1)). * $p < 0.05$, ** $p < 0.01$, n.s. non-significant. Statistical significance was assessed using Kruskal-Wallis ANOVA on ranks, unpaired Student t -test or the non-parametric Mann-Whitney rank sum test.

